

Solution 1. First we will prove that $x = \sum_{P \in \mathcal{P}} \mu_P \mathbf{1}_P + \sum_{C \in \mathcal{C}} \lambda_C \mathbf{1}_C$, where λ and μ are the values returned by Decompose-Flow function. By line 3 of the algorithm, we have that $x = x^+ \upharpoonright_{A(D)}$. Using theorem 44, we have that $x = (\sum_{C \in \mathcal{C}(D^+)} \lambda_C^+ \mathbf{1}_C) \upharpoonright_{A(D)} = \sum_{C \in \mathcal{C}(D^+)} \lambda_C^+ \mathbf{1}_C \upharpoonright_{A(D)}$. Expanding the somation considering the circuits that are in D and the ones that are only in D^+ , we have that $x = \sum_{C \in \mathcal{C}(D)} \lambda_C^+ \mathbf{1}_C \upharpoonright_{A(D)} + \sum_{C \in \mathcal{C}(D^+) \setminus \mathcal{C}(D)} \lambda_C^+ \mathbf{1}_C \upharpoonright_{A(D)}$. By lines 5 and 9 of the algorithm and rewriting some terms, we have that $x = \sum_{C \in \text{supp}(\lambda^+) \setminus \mathcal{P}'} \lambda_C^+ \mathbf{1}_C \upharpoonright_{A(D)} + \sum_{C \in \mathcal{P}'} \lambda_C^+ \mathbf{1}_C \upharpoonright_{A(D)}$. Finally, by lines 5, 9 and 10, we have that $x = \sum_{C \in \mathcal{C}} \lambda_C \mathbf{1}_C + \sum_{P \in \mathcal{P}} \mu_P \mathbf{1}_P$.

Now we will prove that $\text{val}(x) = \mathbf{1}^\top \mu$. By line 3 of the algorithm, we have that $\text{val}(x) = x_{a^+}^+ = e_{a^+}^\top x^+$. Expanding the expression, we have that $\text{val}(x) = e_{a^+}^\top \sum_{C \in \mathcal{C}(D^+)} \lambda_C^+ \mathbf{1}_C$. Then, we have that $\text{val}(x) = \sum_{C \in \mathcal{C}(D^+)} \lambda_C^+ (e_{a^+}^\top \mathbf{1}_C) = \sum_{C \in \mathcal{C}(D^+) \setminus \mathcal{C}(D)} \lambda_C^+$. By line 9 of the algorithm, we have that $\text{val}(x) = \mathbf{1}^\top \mu$.

Finally, we will prove that $|\text{supp}(\mu)| + |\text{supp}(\lambda)| \leq |\text{supp}(x)| + 1$. First, we have that $|\text{supp}(\lambda^+)| = |\text{supp}(\lambda^+ \upharpoonright_{\text{supp}(\lambda^+) \setminus \mathcal{P}'})| + |\text{supp}(\lambda^+ \upharpoonright_{\mathcal{P}'})| = |\text{supp}(\lambda)| + |\text{supp}(\mu)|$. Also, we have that $|\text{supp}(x^+)| = |\text{supp}(x)| + [a^+ \in \text{supp}(x^+)] \leq |\text{supp}(x)| + 1$. By theorem 44, we have that $|\text{supp}(\lambda^+)| \leq |\text{supp}(x^+)|$. Using that and what was established before, we have that $|\text{supp}(\lambda)| + |\text{supp}(\mu)| = |\text{supp}(\lambda^+)| \leq |\text{supp}(x^+)| \leq |\text{supp}(x)| + 1$. Then follows that $|\text{supp}(\lambda)| + |\text{supp}(\mu)| \leq |\text{supp}(x)| + 1$.

Solution 2. First, we will describe the instance of the min-cost flow problem that Min-Cost-Flow-Inst-From-Max-Flow-Inst returns. The auxiliary digraph $D' := (V, A', \varphi')$ is built from D by adding an arc $rs \notin A(D)$ from r to s . Then, we set $\varphi'(a) = \varphi(a)$ for $a \in A(D)$ and $\varphi'(rs) = rs$. Also, we have that $u' \in \mathbb{R}_+^A$ is such that $u'(a) = u(a)$ for $a \in A(D)$ and $u'(rs) = u(\delta_D^{\text{out}}(\{r\}))$. In respect to c' , we have that $c' \in \mathbb{R}^A$ is such that $c'(a) = 0$ for $a \in A(D)$ and $c'(rs) = 1$. It is important mentioning that the r' returned is r and s' is s . To conclude the description, we have $\phi' \in \mathbb{R}_+$ equal to $\delta_D^{\text{out}}(\{r\})$, an upperbound to the maximum value of a rs -flow in D . Given that, our Min-Cost-Flow instance is a feasible instance. It is worth writing that ϕ' smaller or equal to the maximum value of a rs -flow in D' because of $u'(rs)$.

Regarding the Post-Process, it returns $x := x' \upharpoonright_{A(D)}$. Now we will prove that x is a maximum rs -flow in D . First, we know that Min-Cost-Flow in line 3 returns a minimum cost $r's'$ -flow in D' with flow value equal to ϕ' . As the cost is given by $c'^T x'$, by our definitions, the cost is equal to $x_{r's'}$. Also, we know that $\text{val}(x) = \text{val}(x') - x_{r's'}$. By the argument of the Min-Cost-Flow function, we have that $\text{val}(x) = \delta_D^{\text{out}}(\{r\}) - x_{r's'}$. If there is an rs -flow y in D such that $\text{val}(y) > \text{val}(x)$ and the Post-Process algorithm returns x , then it is a contradiction because it means that Min-Cost-Flow function returned x' instead of y' (abusing notation without without harming understanding), but we have that $y_{r's'} < x_{r's'}$ (by the equation of the value of a flow). Thus, x is a maximum value rs -flow in D . It is worth mentioning that, as we use an upperbound for the rs -flow in D , then all possible flows are allowed in our description.

Solution 3. Below is the pseudocode requested. The function Circulation-Inst-from-Bounded-Transshipment(D, u, b) returns an auxiliary digraph $D' := (V \cup \{r, s\}, A', \varphi')$ created as follows. Set $R^- := \{v \in V : b_v < 0\}$, set $R^+ := \{v \in V : b_v \geq 0\}$ and set $S^+ := \{v \in V : b_v > 0\}$. I added new vertices $r, s \notin V(D)$, and set $A' := A \cup R^- \cup R^+ \cup S^+ \cup \{sr\}$, where we may assume that $\{A, R^-, R^+, S^+, sr\}$ is pairwise disjoint (by relabeling common elements), where the incidence function φ' , arc capacities $u' : A' \rightarrow \mathbb{R}_+$ and the vector $l \leq u'$ are:

- $\varphi'(v) := rv, l_v := -b_v, u'(v) := \infty$, for each $v \in R^-$
- $\varphi'(v) := rv, l_v := 0, u'(v) := \infty$, for each $v \in R^+$
- $\varphi'(v) := vs, l_v := 0, u'(v) := b_v$, for each $v \in S^+$
- $\varphi'(sr) := sr, l_{sr} := 0, u'(sr) := \infty$
- $\varphi'(a) := \varphi(a), l_a := 0, u'(a) := u(a)$, for each $a \in A$

Algorithm 1 Bounded-Transshipment(D, u, b)

- 1: $(D', u', l) := \text{Circulation-Inst-from-Bounded-Transshipment}(D, u, b)$
 - 2: $(\text{outcome}, C) := \text{Circulation}(D', u', l)$
 - 3: **if** outcome = feasible **then**
 - 4: **return** (feasible, $C \upharpoonright_{A(D)}$)
 - 5: **else if** outcome = infeasible **then**
 - 6: **return** (infeasible, $V \setminus C$)
 - 7: **end if**
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First, we will prove that the algorithm returns a vector $x \in \mathbb{R}_+^A$ such that $\text{excess}_x \leq b$ and $x \leq u$ when the outcome is feasible. This is the scenario which the algorithm reach line 4 and thus C is a circulation x' in D' such that $l \leq x' \leq u'$. First, we have that $x \leq u$ because $x' \leq u'$ and $u'_a = u_a$ for every $a \in A(D)$. Second, because we made a call to Circulation, we have that $\text{excess}_{x'}(v) = 0$ (in D') for every $v \in V(D')$. As we are returning $x := x' \upharpoonright_{A(D)}$, we have that $\text{excess}_x(v)$ (in D), for an arbitrary $v \in V(D)$, is equal to $x(\delta_{D'}^{\text{out}}(v) \setminus \delta_D^{\text{out}}(v)) - x(\delta_{D'}^{\text{in}}(v) \setminus \delta_D^{\text{in}}(v))$. Given that, we will prove that $\text{excess}_x \leq b$ considering that $l_a \leq x_a \leq u'_a$ for $a \in A(D)$. For an arbitrary $v \in \{v \in V(D) : b_v < 0\}$, we have that $\text{excess}_x(v) = -x_v$, thus $l_v \leq -\text{excess}_x(v) \leq u'_v$, which is the same as $b_v \geq \text{excess}_x(v) \geq -\infty$. For an arbitrary $v \in \{v \in V(D) : b_v = 0\}$, we have that $\text{excess}_x(v) = -x_v$, thus $l_v \leq -\text{excess}_x(v) \leq u'_v$, which is the same as $b_v \geq \text{excess}_x(v) \geq -\infty$. Finally, for an arbitrary $v \in \{v \in V(D) : b_v > 0\}$, we have that $\text{excess}_x(v) = x_{vs} - x_{rv}$, where $\varphi'(vs) = vs$ and $\varphi'(rv) = rv$. This also gives that $b_v \geq \text{excess}_x(v) \geq -\infty$. We proved that x satisfies all the necessary conditions.

Finally, on the infeasible scenario (line 6), we have that C is such that $C \subseteq V(D')$ and $l(\delta_{D'}^{\text{out}}(C)) > u'(\delta_D^{\text{in}}(C))$. By this inequality, we have that $\{r, s\} \subseteq C$. Given that, we have $-b(\{v \in R^-\} \cap \{v \notin C\}) > u(\delta_D^{\text{in}}(C)) + b(\{v \in R^+\} \cap \{v \notin C\})$. Then, we have that $b(\{v \notin C\} \cap \{v \in V\}) + u(\delta_D^{\text{in}}(C)) < 0$. Finally, this is the same as $b(V \setminus C) + u(\delta_D^{\text{out}}(V \setminus C)) < 0$. Thus, we also return the correct output in the infeasible scenario.

Solution 4. Below is the pseudocode requested. Set $V_1 := \{v_1 \in V\}$ and set $V_2 := \{v_2 \in V\}$. The function $\text{Max-Flow-Inst-from-Common-Transversal}(\{r, s\} \cup \mathcal{H}_1, \mathcal{H}_2)$ returns the digraph $D := (\mathcal{E}_1 \cup V_1 \cup V_2 \cup \mathcal{E}_2 \cup, A, \varphi)$ created as follows.

Set $A_1 := \{re : e \in \mathcal{E}_1\}$, set $A_2 := \{ev_1 : e \in \mathcal{E}_1, v_1 \in V_1, v_1 \subseteq e\}$, set $A_3 := \{v_1v_2 : v_1 \in V_1, v_2 \in V_2, v_1 = v_2\}$, set $A_4 := \{v_2e : e \in \mathcal{E}_2, v_2 \in V_2, v_2 \subseteq e\}$ and set $A_5 := \{es : e \in \mathcal{E}_2\}$. Then, set $A := A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5$, where we may assume that $\{A_1, A_2, A_3, A_4, A_5\}$ is pairwise disjoint (by possibly relabeling common elements), where the incidence function φ and arc capacities $u : A \rightarrow \mathbb{R}_+$ are:

- $\varphi(re) := re, u(re) := 1$, for each $re \in A_1$
- $\varphi(ev_1) := ev_1, u(ev_1) := \infty$, for each $ev_1 \in A_2$
- $\varphi(v_1v_2) := v_1v_2, u(v_1v_2) := 1$, for each $v_1v_2 \in A_3$
- $\varphi(v_2e) := v_2e, u(v_2e) := \infty$, for each $v_2e \in A_4$
- $\varphi(es) := es, u(es) := 1$, for each $es \in A_5$

It is important mentioning that MaxFlow-MinCut never returns the infeasible case because of the way we build D .

Algorithm 2 Common-Transversal(D, u, b)

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1:  $(D, u, r, s) := \text{MaxFlow-MinCut-Inst-from-Common-Transversal}(\mathcal{H}_1, \mathcal{H}_2)$ 
2:  $(\_, (x, R)) := \text{MaxFlow-MinCut}(D, u, r, s)$ 
3: if  $\text{val}(x) = u(\delta_D^{\text{in}}(\{s\}))$  then
4:    $(\mu, \lambda) := \text{Decompose-Flow}(D, r, s,)$ 
5:    $T := \emptyset$ 
6:   for each  $P \in \text{supp}(\mu)$  do
7:      $P := \langle v_0, a_1, v_1, a_2, v_2, \dots, v_l \rangle$ 
8:      $T \leftarrow T \cup \{v_2\}$ 
9:   end for
10:  return (feasible,  $T$ )
11: else if  $\text{val}(x) < u(\delta_D^{\text{in}}(\{s\}))$  then
12:  return (infeasible,  $(\mathcal{E}_1 \cap R, \mathcal{E}_2 \setminus R)$ )
13: end if

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First, we will prove that the algorithm returns a common transversal when the outcome is feasible. This is the scenario which the algorithm reach line 10. In this case, all the entries in x are integral, in fact they are in $\{0, 1\}$. This holds even in this case that we have infinity capacities because if $\delta_D^{\text{out}}(\{v\})$ has an arc with infinity capacity, then every arc in $\delta_D^{\text{in}}(\{v\})$ has capacity equal to 1. Given this and considering how the arcs in A_2 are built, we can affirm that in the feasible case we have exactly m disjoint rs -paths in $\text{supp}(\mu)$. Also, $\text{supp}(\lambda)$ is the empty set because we do not have circuits in D . By that, each one of those paths has a unique $e \in \mathcal{E}_1$, a unique $e \in \mathcal{E}_2$ and a unique $v \in V$ such that $v \subseteq \mathcal{E}_1$ and $v \subseteq \mathcal{E}_2$. Given that the algorithm returns the set of all of those vertices, we return a common transversal.

Finally, on the infeasible scenario (line 12), we have that $u(\delta_D^{\text{out}}(R)) < u(\delta_D^{\text{in}}(\{s\}))$, which is the same as $|\delta_D^{\text{out}}(R)| < m$. Expanding the expression, we have that $|\delta_D^{\text{out}}(R) \cap A_1| + |\delta_D^{\text{out}}(R) \cap A_3| + |\delta_D^{\text{out}}(R) \cap A_5| < m$. Then we have that $|\mathcal{E}_1 \setminus R| + |(\bigcup (\mathcal{E}_1 \cap R)) \cap (\bigcup (\mathcal{E}_2 \setminus R))| + |\mathcal{E}_2 \cap R| < m$. Finally, we have that $|(\bigcup (\mathcal{E}_1 \cap R)) \cap (\bigcup (\mathcal{E}_2 \setminus R))| < |\mathcal{E}_1 \cap R| + |\mathcal{E}_2 \setminus R| - m$. Thus, we also return the correct output in the infeasible scenario.